

Proposal for CIE6032 Final Project 2026

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Title: Screen Capture Rectification and Temporal Stabilization for Real-world Captured-screen Videos

Description: Existing screen demoiréing and screen-image restoration studies show that captured-screen content is a meaningful image processing problem. However, many public datasets are collected in controlled settings, where the screen region is already cropped, aligned, or paired with a clean reference. In real phone-captured videos, the input usually contains background regions, perspective distortion, hand-held camera shake, weak or missing screen borders, moiré patterns, glare, and dynamic screen content. This project focuses on this missing preprocessing stage: converting a full captured-screen video into a rectified and temporally stable screen-coordinate video before downstream demoiréing, OCR, or archival restoration.

Task and goal: The input is a hand-held video of a computer screen. The output is a fixed-ratio, front-facing, temporally stable video of the screen content. The goal is to remove most non-screen background, correct perspective distortion, reduce frame-to-frame jitter, and provide a reliable normalized input for later restoration tasks.

Method: The proposed pipeline first obtains the first-frame screen quadrilateral by automatic detection or manual annotation. A homography is then used to rectify the screen plane to a 16:9 output canvas. To avoid the jitter caused by independent frame-wise corner detection, reference-plane feature points are tracked using Lucas-Kanade optical flow, and a robust homography is estimated with RANSAC. Unreliable updates are rejected by geometric constraints, while missing or unstable trajectory segments are interpolated and smoothed before rendering the normalized video.

Dataset and experiment: The final evaluation will use a self-collected test set with 5 scenario classes, 10 clips per class, and about 5 seconds per clip. The classes are static webpages/documents, scrolling webpages, in-screen video playback, PPT or weak-border low-texture pages, and 4K/moiré/glare hard cases. Public demoiréing datasets will be discussed as related-work evidence because they mostly evaluate restoration after alignment, while this project evaluates the preceding video rectification and stabilization step.

Evaluation metrics: The main quantitative metrics are residual adjacent-frame translation p95, rotation p95, and scale-delta p95 measured in the normalized output video. Tracker accept ratio, RANSAC inlier count, inlier ratio, and feature coverage will be used to explain failure cases. In an initial 317-frame local test video, reference-plane tracking reduced the last-two-second residual motion to 0.118 px translation p95 and 0.0044 deg rotation p95, compared with 1.927 px and 0.0425 deg for frame-wise detection.

Expected results: The expected result is a classical image-processing and geometric-vision pipeline that produces stable, rectified screen videos from realistic captured-screen inputs. The final report will compare frame-wise detection, ordinary optical-flow tracking, and reference-plane tracking; summarize successful and failed cases; and discuss when this preprocessing is sufficient for downstream screen video restoration.

Tentative timeline/to-do list: Jun. 22–24: finalize proposal and presentation. Jun. 25–30: complete the 50-clip self-collected test set. Jul. 1–7: run method ablations and metric evaluation. Jul. 8–12: prepare visual comparisons, failure analysis, and final report. Jul. 13–15: finalize code, sample data, and presentation.